Yifan Yu

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EDUCATION

University of Illinois Urbana-Champaign, Urbana IL **08/2024 - 05/2028**

Doctor of Philosophy | Computer Science

Advisor: Fan Lai

Research Keywords: Diffusion Language Models, GenAI Systems, RL training, Muti-agent Systems, Model Compression

Research Impact: I-DLM is being adopted in Together.AI and Astera Holdings; IC-Cache is being adopted inside Google; LoftQ is integrated in PEFT and LLaMA-Factory library

Georgia Institute of Technology, Atlanta GA **08/2020 - 05/2024**

Bachelor of Science | Computer Science

Honors: Faculty Honor, Presidential Undergraduate Research Award

RESEARCH INTERESTS

My research interests primarily focus on efficient machine learning, particularly efficient training and inference. Recently, I am interested in:

• Diffusion Language Modes; Multi-Agent Systems; Reinforcement Learning; Large Language Models; Inference Systems

PUBLICATION

5 accepted first/cofirst-author papers *equal contribution

Introspective Diffusion Language Models
Yifan Yu*, Yuqing Jian*, Junxiong Wang, Zhongzhu Zhou, Donglin Zhuang, Xinyu Fang, Sri Yanamandra, Xiaoxia Wu, Qingyang Wu, Shuaiwen Leon Song, Tri Dao, Ben Athiwaratkun, James Zou, Fan Lai, Chenfeng Xu
In submission to Conference On Language Modeling, 2026

CORRECT: COndensed eRror RECognition via knowledge Transfer in multi-agent systems
Yifan Yu, Moyan Li, Shaoyuan Xu, Jinmiao Fu, Xinhai Hou, Fan Lai, Bryan Wang
Accepted by International Conference on Machine Learning (ICML), 2026

XRPO: Pushing the limits of GRPO with Targeted Exploration and Exploitation
Udbhav Bamba*, Minghao Fang*, **Yifan Yu***, Haizhong Zheng, Fan Lai
Accepted by International Conference on Machine Learning (ICML), 2026

IC-Cache: Efficient Large Language Model Serving via In-context Caching
Yifan Yu*, Yu Gan*, Nikhil Sarda, Lillian Tsai, Jiaming Shen, Yanqi Zhou, Arvind Krishnamurthy, Fan Lai, Henry M. Levy, David Culler
Accepted by The 31st Symposium on Operating Systems Principles (S0SP), 2025

LoftQ: LoRA-Fine-Tuning-Aware Quantization for Large Language Models
Yixiao Li*, **Yifan Yu***, Chen Liang, Pengcheng He, Nikos Karampatziakis, Weizhu Chen, Tuo Zhao
Accepted by International Conference on Learning Representations (ICLR), 2024, ***Oral Presentation***

LoSpase: Structured Compression of Large Language Models based on Low-Rank and Sparse Approximations
Yixiao Li*, **Yifan Yu***, Qingru Zhang, Chen Liang, Pengcheng He, Weizhu Chen, Tuo Zhao
Accepted by International Conference on Machine Learning (ICML), 2023

OPPO: accelerating ppo-based rlhf via pipeline overlap
Kaizhuo Yan*, Yingjie Yu*, **Yifan Yu**, Haizhong Zheng, Fan Lai
Accepted by International Conference on Learning Representations (ICLR), 2026

Single-agent or Multi-agent Systems? Why Not Both?
Mingyan Gao*, Yanzi Li*, Banruo Liu*, **Yifan Yu**, Phillip Wang, Ching-Yu Lin, Fan Lai
In submission to ACL Rolling Review

WORK EXPERIENCE

Part-time Research Intern in Together.AI **01/2026 – Now**

- Worked on introspective diffusion language models, the first diffusion language model to achieve on-par performance with autoregressive thinking models
- Implemented training based on llamafactory; designed and implemented inference using SGLang to achieve 3.1-4x throughput gains over LlaDa-2.1 series models and SDAR series models

Applied Scientist Intern in Amazon **05/2025 – 08/2025**

- Worked in automatic error attribution for multi-agent systems.
- Proposed a thought-template based method that can significantly improve the error detection accuracies; synthesizing datasets based on Magnetic One to support further research on automatic error attributions.

Student Researcher in Systems Research@Google **05/2024 – 07/2024**

- Worked in efficient systems for serving LLM facing millions of requests leveraging the abundance of users' queries.
- Implemented experiments on millions of real users' requests using Huggingface and vLLM backends; observed significant quality and throughput improvements on Alpaca, Open Orca, and LMSys-chat.

Intern Data Scientist in The Home Depot **05/2022 – 07/2022**

- Worked in the online search team with DSI-based retrieval model as the first undergraduate ever in the team.
- Implemented DSI with PyTorch and HuggingFace; proposed an encoding method based on Pecos clustering for the DSI indexing process; improved over 15 percent recall rate over baseline, matching the current production search performance.

RESEARCH

Research Assistant at AISys Lab. **08/2024 – Present**

- Worked on efficient serving of LLM facing substantial numbers of requests advised by Prof. Fan Lai
- Implemented a system prototype that can significantly improve the latency and throughput of LLM servings as presented in IC-Cache paper leveraging the in-context learning theory; supported vllm, Gemini backend, and Langchain backends.

Research Assistant at FLASH (Foundations of LeArning Systems for Alchemy) **10/2021 – 05/2024**

- Worked on improving the efficiencies of large language models with Prof. Tuo Zhao.
- Combined the pruning methods and low rank decomposition as presented in LoSparse paper; implemented the codes and conducted the experiments with Huggingface and PyTorch
- Proposed and implemented an alternative optimization method to minimize the quantization error as presented in the LoftQ paper; surpassed QLoRA on different precisions; adopted by PEFT and LLaMA-Factory

SKILLS

Programming: Python (Tensorflow2, Pytorch), Java, C, HTML, CSS, JavaScript, SQL, MATLAB, Mathematica
Related course: GenAI system, Text mining for LLM, Machine Learning, Computer Organization and Programming,